

MATE 6551: Tarea 3

Due on December 13, 2025

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Problem 0

Let $A \subseteq X$ and $B \subseteq Y$ be given. Assume that $f_0, f_1 : X \rightarrow Y$ with $f_0|_A = f_1|_A$ and $f_i(A) \subseteq B$ for $i \in \{0, 1\}$; assume $g_0, g_1 : Y \rightarrow Z$ with $g_0|_B = g_1|_B$. If $f_0 \simeq f_1 \text{ rel } A$ and $g_0 \simeq g_1 \text{ rel } B$, then $g_0 \circ f_0 \simeq g_1 \circ f_1 \text{ rel } A$.

Proof:

Let $H_1 : X \times I \rightarrow Y$ be a homotopy $f_0 \simeq f_1 \text{ rel } A$ and $H_2 : Y \times I \rightarrow Z$ be a homotopy $g_0 \simeq g_1 \text{ rel } B$. Now define $H : X \times I \rightarrow Z$ by $H(x, t) = H_2(H_1(x, t), t)$. Note that this is a composition of continuous functions and thus continuous. Furthermore, note that

$$H(x, 0) = H_2(H_1(x, 0), 0) = g_0(f_0(x)) = (g_0 \circ f_0)(x) \quad (1)$$

and

$$H(x, 1) = H_2(H_1(x, 1), 1) = g_1(f_1(x)) = (g_1 \circ f_1)(x) \quad (2)$$

This shows that H is a homotopy, to show that it is relative to A , we use that H_1 is a homotopy rel A and H_2 is a homotopy rel B . For $a \in A$ and any $t \in I$, we have $H(a, t) = H_2(H_1(a, t), t) = H_2(f_0(a), t)$, but $f_0(A) \subseteq B$ yields:

$$H(a, t) = g_0(f_0(a)) = (g_0 \circ f_0)(a) \quad (3)$$

Similarly, we have:

$$H(a, t) = H_2(H_1(a, t), t) = H_2(f_1(a), t) = g_1(f_1(a)) = (g_1 \circ f_1)(a) \quad (4)$$

$\therefore H$ is a homotopy $g_0 \circ f_0 \simeq g_1 \circ f_1 \text{ rel } A$.

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Problem 1

Let $f_0 \simeq f_1 \text{ rel } \dot{I}$ and $g_0 \simeq g_1 \text{ rel } \dot{I}$ be paths in X and Y respectively. If for $i \in \{0, 1\}$, (f_i, g_i) is the path in $X \times Y$ defined by $t \mapsto (f_i(t), g_i(t))$. Prove that $(f_0, g_0) \simeq (f_1, g_1) \text{ rel } \dot{I}$.

Proof:

Let $H_1 : I \times I \rightarrow X$ be a homotopy $f_0 \simeq f_1 \text{ rel } \dot{I}$, and $H_2 : I \times I \rightarrow Y$ be a homotopy $g_0 \simeq g_1 \text{ rel } \dot{I}$. Then define $H : I \times I \rightarrow X \times Y$ by $H(t, s) = (H_1(t, s), H_2(t, s))$. This is clearly continuous. Note that

$$H(t, 0) = (H_1(t, 0), H_2(t, 0)) = (f_0(t), g_0(t)) = (f_0, g_0)(t) \quad (5)$$

and

$$H(t, 1) = (H_1(t, 1), H_2(t, 1)) = (f_1(t), g_1(t)) = (f_1, g_1)(t) \quad (6)$$

Finally, for $i \in \dot{I}$ and any $s \in I$,

$$H(i, s) = (H_1(i, s), H_2(i, s)) = (f_0(i), g_0(i)) = (f_0, g_0)(i) \quad (7)$$

and

$$H(i, s) = (H_1(i, s), H_2(i, s)) = (f_1(i), g_1(i)) = (f_1, g_1)(i) \quad (8)$$

$\therefore H$ is a homotopy $(f_0, g_0) \simeq (f_1, g_1) \text{ rel } \dot{I}$.

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Problem 2

If $\pi_1(Y, y_0) \neq \{1\}$ for some pointed space (Y, y_0) , then $\pi_1(S^1, 1) \neq \{1\}$.

Proof:

Suppose $\pi_1(S^1, 1) = \{1\}$, then 1_{S^1} is nullhomotopic, since there is only one path homotopy class, so $[1_{S^1}] = [k]$ where $k(z) = 1$. Let $f : I \rightarrow Y$ be a closed path at y_0 . Then, since $S^1 \xrightarrow{\eta} I/\dot{I}$, and $f(0) = f(1) = y_0$, the composite map $f\pi^{-1}\eta : S^1 \rightarrow Y$, where $\pi : I \rightarrow I/\dot{I}$ is the quotient projection, is well defined and continuous. Of course, π^{-1} itself is not well defined, we can use it in our expression only because f is constant on $\dot{I}$. Now, using the fact that relative homotopy is a congruence in **Top**, we have:

$$\begin{aligned} 1_{S^1} &\simeq k \\ \implies f\pi^{-1}\eta 1_{S^1} &\simeq f\pi^{-1}\eta k \\ \implies f &\simeq f\pi^{-1}\eta k\eta^{-1}\pi \end{aligned} \quad (9)$$

But k is constant, so $f\pi^{-1}\eta k\eta^{-1}\pi$ must be constant too. The result follows by contrapositive.

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Problem 3

Prove that S^1 is not a retract of D^2 .

Proof:

Suppose, for a contradiction, that S^1 is a retract of D^2 . Then there exists a retraction $r : D^2 \rightarrow S^1$. Therefore, the following diagram commutes, where $i : S^1 \rightarrow D^2$ is the inclusion:

$$\begin{array}{ccc} S^1 & \xrightarrow{1_{S^1}} & S^1 \\ & \searrow i & \nearrow r \\ & D^2 & \end{array}$$

Applying the fundamental group functor, we get:

$$\begin{array}{ccc}
 \pi_1(S^1) = \mathbb{Z} & \xrightarrow{\pi_1(1_{S^1}) = 1_{\pi_1(S^1)} = 1_{\mathbb{Z}}} & \pi_1(S^1) = \mathbb{Z} \\
 & \searrow \pi_1(i) \quad \nearrow \pi_1(r) & \\
 & \pi_1(D^2) = 0 &
 \end{array}$$

which has to commute by functoriality of π_1 . But then, since $\pi_1(D^2) = 0$,

$$1_{\mathbb{Z}} = \pi_1(r) \circ \pi_1(i) = 0 \quad (10)$$

which is a contradiction. ✖

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Problem 4

Let f be a closed path in S^1 at 1. Prove that if f is not surjective, then $\deg f = 0$. Give an example of a surjective f with $\deg f = 0$.

Proof:

Let $\tilde{f} : I \rightarrow \mathbb{R}$ denote the unique lifting of f with $\tilde{f}(0) = 0$. Suppose that $\deg f \neq 0$, in fact, without loss of generality suppose that $\deg f > 0$. Because $\tilde{f}$ is continuous, and $\deg f \in \mathbb{Z}$, we have that the image $\tilde{f}(I) = [\tilde{f}(0), \tilde{f}(1)] = [0, \deg f]$. But $\exp \circ \tilde{f} = f$, so we have:

$$f(I) = (\exp \circ \tilde{f})(I) = \exp(\tilde{f}(I)) = \exp([0, \deg f]) \quad (11)$$

But $\deg f \in \mathbb{Z}$ and $\deg f > 0 \implies [0, 1] \subseteq [0, \deg f] \implies \exp([0, 1]) \subseteq \exp([0, \deg f])$. But $\exp([0, 1]) = S^1$ by construction, so $S^1 = \exp([0, 1]) \subseteq \exp([0, \deg f]) = f(I)$.

$\therefore f$ is surjective.

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